CIDROY INFOTECH PRIVATE LIMITED | cidroy Infotech

Senior Technical Architect & Engineering Lead (Backend / Distributed Systems)

8–12+ Years Experience

About Cidroy

Cidroy builds mission-critical software for enterprises, industrial, and high-security environments, spanning enterprise platforms, real-time monitoring systems, and complex data-intensive applications where correctness and reliability aren't optional.

The Role

You will be the technical architect and engineering lead for a large-scale water infrastructure monitoring platform that tracks water from source through city zones and reservoirs to consumers, built on live SCADA telemetry (flow, pressure, quality, geolocation) ingested at scale. You will own the architecture end to end: distributed systems design, data modelling for a deep zone hierarchy, high-availability and failover planning, and the integrations that bring third-party SCADA data in reliably. You will lead a small engineering team while staying hands-on, writing code, reviewing designs, and driving decisions from concept to production. This is not an architect-from-a-distance role; you will be as comfortable in an infrastructure review as you are in production code.

What You Will Own

- Architecture and delivery of scalable backend systems, APIs, and event-driven pipelines using appropriate technologies and architectural patterns for high-frequency SCADA telemetry across a citywide zone hierarchy
- Data modelling and system design for reliable aggregation of leakage, pressure and Non-Revenue Water (NRW) metrics as data volume grows
- Engineering leadership for a small team, spanning architecture, development and delivery
- System reliability: high availability, failover planning, observability and scalability across the platform
- Integrations with third-party SCADA systems, industrial platforms and operational data sources
- Technical and infrastructure design reviews, catching expensive mistakes before they are built
- Direct work with stakeholders to translate ambiguous, evolving requirements into working systems

Must-Have

Architecture & Leadership

- Experience building and delivering production-grade software systems at scale
- Strong backend architecture and distributed systems experience
- Experience leading engineering teams while remaining hands-on with development

Core Technical Expertise

- Strong hands-on backend development experience in JavaScript/TypeScript, Java, or another relevant backend language, with the ability to work comfortably at code level
- Deep backend architecture and distributed systems experience, regardless of programming language
- Strong PostgreSQL knowledge, including performance tuning and data modelling

- Experience with Kafka or similar event-streaming platforms
- Experience with TimescaleDB, PostGIS and GIS or geospatial technologies

Language-agnostic role: We are open to strong backend engineers from JavaScript/TypeScript, Java, or other relevant backend backgrounds. What matters most is depth in backend architecture, distributed systems, scalability, reliability, and hands-on engineering.

Integrations & Operations

- Experience integrating with SCADA systems, OPC UA, MQTT, Industrial IoT or telemetry platforms
- Working knowledge of Docker, Kubernetes or K3s, Prometheus and Grafana
- Experience designing highly available, fault-tolerant systems, including capacity planning and infrastructure reviews

Good to Have

- GeoServer, MapLibre or related geospatial platforms
- Keycloak, Kong, API gateways and secure platform design
- Exposure to mobile architecture (Flutter), analytics, ML-driven systems, or other mission-critical platforms

What Strong Looks Like Here

- You would rather be handed a problem than a fully-defined ticket, and you learn unfamiliar domains quickly
- You look at a hierarchy like City > Zone > Sub-zone > Reservoir and design for it to stay fast and reliable as data volume grows, not just at demo scale
- You move comfortably between architecture discussions, infrastructure reviews, integrations and production code, regardless of the language or framework you have primarily worked with.
- You can spot an expensive mistake before it is built, not after it ships

Who This Is Not For

- Architects who prefer to design from a distance and hand off implementation entirely
- Candidates without experience leading a team while remaining hands-on with code
- Candidates who have not worked with distributed, high-availability systems in production
- Candidates whose experience is primarily language/framework-specific without strong transferable backend architecture and distributed-systems experience

Why Cidroy

Cidroy builds software for environments where failure is not an option: defence-adjacent, industrial, and high-security deployments across India. You will be leading architecture and engineering for a platform that tracks water across an entire city network and helps pinpoint exactly where it is being lost. Direct access to the founding team, real ownership, and problems that matter.